

Yuni Wu

South San Francisco, CA

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Summary

Robotics AI engineer with experience building ROS-based data pipelines, VLM navigation systems, state-action datasets, and evaluation frameworks for embodied AI and social navigation research.

Technical Skills

Languages: Python, Go, SQL, R

Robotics Data: ROS telemetry, rosbag extraction, robotics data pipelines, perception outputs, state-action datasets

Navigation AI: VLM navigation, behavior cloning, Qwen/InternVL action labeling,

STOP/FORWARD/LEFT/RIGHT/REVIEW decisions

Evaluation: offline evaluation, scenario benchmarking, residual diagnostics, reliability evaluation, inference latency benchmarking

ML & Systems: Transformers, CNNs/ResNet, calibration, cross-validation, Docker, Cloud Run inference evaluation, structured logging

Projects

VLM-Based Social Navigation for Quadruped

[project]

- Developed a hybrid robot navigation system combining behavior cloning with VLM-based decision rules.
- Constructed ROS/rosbag telemetry pipelines to transform raw robot data into structured state-action datasets for offline evaluation.
- Modeled human interaction context using VLMs (Qwen/InternVL) to generate discrete STOP/FORWARD/LEFT/RIGHT/REVIEW actions.
- Built offline evaluation pipelines across curated rosbag scenarios to compare geometry-based and VLM-based decision strategies with sample- and scenario-level analysis.
- Achieved 0.11 MAE and $R^2 = 0.62$ under trajectory-split offline evaluation and identified rotational instability (P99 ≈ 0.80) through residual diagnostics.

NailAI — Cloud-Native AI Inference Platform

- Built event-driven image inference workflows using Cloud Run, Pub/Sub, and Cloud Storage for scalable asynchronous processing.
- Achieved 0% failure at 200 concurrent users through idempotent job orchestration, retry handling, and metadata-driven monitoring workflows.

Experience

Graduate Research Assistant

Sept 2025 – May 2026

CAIRO Lab, University of Colorado Boulder

- Developed robotics data processing pipelines to convert rosbag recordings into structured datasets for embodied AI navigation experiments.
- Established robotics evaluation pipelines to benchmark navigation policies across ROS-based human-interaction scenarios, enabling reproducible experimentation and rapid policy iteration.
- Processed ROS telemetry and perception outputs into state-action datasets for navigation analysis, reliability evaluation, and inference latency benchmarking.

Machine Learning Engineer Intern

May 2025 – Aug 2025

MyEdMaster LLC

- Evaluated ResNet architecture tradeoffs through ablation workflows, reducing inference latency by $5\times$ ($0.15s \rightarrow 0.03s$ per image).
- Designed evaluation pipelines with cross-validation, ROC-AUC, and calibration to measure model reliability under class imbalance.

Software Engineer

June 2021 – Aug 2022

Shopee Co. Ltd.

- Engineered validation and anomaly detection systems across 800+ production scenarios, reducing manual QA workload by 80%.
- Built data quality monitoring pipelines to detect regression anomalies and surface near real-time reporting inconsistencies, accelerating issue detection by 78%.

Education

University of Colorado Boulder, MS, Data Science GPA: 3.76/4.0

May 2026

Soochow University, BS, Data Science GPA: 3.7/4.0

June 2021